

Project Deliverables & Scope of Work

Geospatial Analysis – Habitat Suitability & Agricultural Analytics

1. Project Overview

Project Attribute	Details
PO Number	PO-DS-AGRI-2026-27-01
PO Date	22 May 2026
Project Title	Geospatial Analysis – Habitat Suitability & Agricultural Analytics
End Client	Meghalaya Farmers' Empowerment Commission Development (MFEC) Government of Meghalaya
SOW Reference	MFEC Buckwheat and Fruit Wine Initiatives – Meghalaya
Contract Validity	Until 31 August 2026

2. Project Objective

The objective of this project is to support the MFEC Buckwheat and Fruit Wine Initiatives in Meghalaya through advanced geospatial analysis, habitat suitability modelling, agricultural intelligence, climate-risk assessment, and GIS-based decision-support tools.

The project aims to identify optimal cultivation zones, assess crop health and environmental suitability, and provide actionable insights for agricultural expansion and planning.

3. Detailed Scope of Work & Deliverables

Crop Suitability & Core Modelling

Deliverable 1: Habitat Suitability Modelling – Buckwheat

- Develop MaxEnt habitat suitability model for Buckwheat across 9 districts of Meghalaya.
 - Generate suitability classifications: High Suitability, Medium Suitability, and Low Suitability.
 - Conduct terrain typology analysis including Elevation, Slope, and Aspect.
 - Rank Top 5 priority blocks in each district ranked by suitability potential.
- Outputs:** GeoTIFF raster layers, high-resolution PDF suitability maps exports, and district-wise priority ranking tables.

Deliverable 2: Habitat Suitability Modelling – Wine Fruits

- Develop separate MaxEnt models for: Plum, Peach, and Passion Fruit.
- Overlay terrain characteristics.
- Identify priority cultivation zones.

Outputs: GeoTIFF rasters, PDF map reports, and priority zone analysis.

Deliverable 3: Model Validation Report

- Conduct AUC (Area Under Curve) validation for all crop models.
- Assess environmental variable contribution.
- Validate model outputs against field survey observations.

Acceptance Criteria: Target AUC Score: > 0.70

Outputs: Model Validation Report, Predictor Importance Analysis, and Ground Truth Validation Findings.

Satellite Analytics & Environmental Risk

Deliverable 4: Crop Presence and Area Mapping

- Satellite-based identification of current buckwheat cultivation areas.
- Land-use and land-cover analysis.
- Mapping of surveyed cultivation sites.

Outputs: Vector shapefiles, cultivation distribution maps, and expansion opportunity maps.

Deliverable 5: Crop Health Assessment (NDVI & NDWI)

- NDVI-based vegetation health analysis.
- NDWI-based moisture stress assessment.
- Correlation of vegetation health with field observations.

Data Sources: Sentinel-2 Imagery, Copernicus Program, USGS Repository.

Outputs: Crop Health Maps, Moisture Stress Maps, and Analytical Report.

Deliverable 6: Climate Suitability and Risk Analysis

- Historical climate assessment.
- Frost risk mapping for buckwheat.
- Temperature suitability assessment for wine fruits.

- Rainfall adequacy evaluation.

Data Sources: WorldClim, Terrain and climate datasets.

Outputs: Climate Suitability Maps, Frost Risk Maps, and Rainfall Analysis Report.

Deliverable 7: Water Management Insights

- Crop Water Requirement (CWR) estimation.
- Rainfall adequacy assessment.
- Seasonal water availability analysis.

Outputs: Water Requirement Report, Water Availability Maps, and Rain-fed Cultivation Assessment.

Strategic Reporting & Platforms

Deliverable 8: Priority Zone Report

- Rank Top 5 blocks per district.
- Classify areas into: High Suitability, Medium Suitability, and Low Suitability.
- Estimate area under each suitability class.

Outputs: District Priority Ranking Report, Excel Summary Tables, and Suitability Statistics.

Deliverable 9: Interactive GIS Dashboard (Subject to Budget Approval)

- Web-based GIS Viewer with layer switching capabilities.
- District-level filtering and interactive map queries.
- Download functionality for maps and datasets.

Outputs: GIS Dashboard Deployment and User Access Documentation.

Deliverable 10: Final Deliverable Package

- **Data Deliverables:** GeoTIFF raster datasets, vector shapefiles, district and block boundaries, priority zone layers, cultivation site layers.
- **Reporting Deliverables:** High-resolution PDF maps, Excel summary sheets, analytical reports, suitability rankings.

Deliverable 11: Training and Handover Support

- Walkthrough of analytical outputs and dashboard demonstration (if applicable).
- Methodology explanation sessions and user guide preparation.

- Knowledge transfer and Q&A support.

Outputs: Training Sessions, User Manual, Methodology Documentation, and Handover Completion Report.

4. Project Timeline & Deliverables Schedule

Project Duration: 22 May 2026 – 31 August 2026

S.No.	Deliverable	Key Activities	Outputs	Timeline	Planned Submission
1	Habitat Suitability Modelling – Buckwheat	MaxEnt modelling, terrain analysis, suitability classification, priority block identification	GeoTIFF suitability maps, PDF maps, district-wise rankings	Week 3-5	30 Jun 2026
2	Habitat Suitability Modelling – Wine Fruits	Crop suitability modelling, terrain overlay, priority zone identification	GeoTIFF suitability maps, PDF maps	Week 3-5	30 Jun 2026
3	Model Validation Report	AUC assessment, variable contribution analysis, field data validation	Validation report, accuracy assessment, predictor analysis	Week 6	03 Jul 2026
4	Crop Presence & Area Mapping	Satellite image classification, cultivation mapping, expansion zone identification	Vector shapefiles, cultivation maps, expansion zone maps	Week 7-8	17 Jul 2026
5	Crop Health Assessment (NDVI & NDWI)	Vegetation health assessment, moisture stress analysis, field data correlation	NDVI maps, NDWI maps, health assessment report	Week 9	24 Jul 2026
6	Climate Suitability & Risk Analysis	Temperature, rainfall and frost-risk assessment	Climate suitability maps, risk analysis report	Week 10	31 Jul 2026
7	Water Management Insights	Crop water requirement analysis, rainfall adequacy assessment, water availability mapping	Water requirement report, seasonal water maps	Week 11	07 Aug 2026

S.No.	Deliverable	Key Activities	Outputs	Timeline	Planned Submission
8	Priority Zone Report	District-wise ranking, area estimation, suitability statistics	Priority block report, Excel statistics, suitability summaries	Week 12	14 Aug 2026
9	Interactive GIS Dashboard*	Dashboard development, testing and deployment	Web GIS Dashboard with interactive layers and filters	Week 12-13	21 Aug 2026
10	Final Deliverable Package	Compilation of all reports, maps and datasets	GeoTIFFs, shapefiles, PDF maps, Excel summaries	Week 13-14	25 Aug 2026
11	Training & Handover Support	User training, methodology walkthrough, Q&A sessions	User guide, training materials, handover report	Week 14	31 Aug 2026

5. Milestone Summary

Milestone	Deliverables Covered	Completion Date
Milestone 1 – Data Preparation & Suitability Modelling	Deliverables 1 & 2	30 Jun 2026
Milestone 2 – Model Validation & Mapping	Deliverables 3 & 4	17 Jul 2026
Milestone 3 – Crop Health & Climate Analytics	Deliverables 5, 6 & 7	07 Aug 2026
Milestone 4 – Priority Zone Identification & Dashboard	Deliverables 8 & 9	21 Aug 2026
Milestone 5 – Final Submission & Knowledge Transfer	Deliverables 10 & 11	31 Aug 2026